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Short communication

Comparative efficacy of flubendazole and a commercially available herbal wormer against natural infections of *Ascaridia galli*, *Heterakis gallinarum* and intestinal *Capillaria* spp. in chickens

S. Squires[a,*], M. Fisher[a], O. Gladstone[a], S. Rogerson[a], P. Martin[a], S. Martin[a], H. Lester[a], R. Sygall[b], N. Underwood[b]

[a] *Ridgeway Research, Park Farm Buildings, Park Lane, St. Briavels GL15 6QX, Gloucestershire, UK*
[b] *Janssen-Cilag Ltd., 50-100 Holmers Farm Way, High Wycombe HP12 4EG, Buckinghamshire, UK*

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ABSTRACT

The efficacy of a commercially available flubendazole-based product and a commercially available herbal product were compared against three species of helminth parasites of chickens: *Ascaridia galli*, *Heterakis gallinarum* and *Capillaria* spp. A total of 48 naturally infected chickens were used in the study with 16 birds in each of three treatment groups (untreated control; flubendazole; and a herbal product). One bird from each treatment group was necropsied on Day 0 prior to first treatment to confirm the parasite species present in the birds. Treatments were administered as labelled and the 45 remaining birds were necropsied on Day 12 and worm counts performed. Average worm counts in the two treated groups were compared to the untreated controls to calculate efficacy. Flubendazole (Group A) achieved an overall efficacy of 99.4% for the three parasite species. The herbal product (Group B) achieved efficacies ranging from less than zero to 11.6% for the three parasites, with worm counts not significantly different to the untreated controls. At present, commercially available herbal products claiming anthelmintic properties do not require licencing as veterinary medicinal products (Directive 2004/28/EC: see Article 17 and 33–38) and thus are not required to meet specific efficacy thresholds. Products which do not appear to deliver acceptable anthelmintic efficacy, are obviously a concern from many aspects but specifically from an animal welfare perspective.

1. Introduction

Ascaridia galli, *Heterakis gallinarum* and *Capillaria* spp. are major pathogenic intestinal parasites of chickens; symptoms include listlessness, emaciation and diarrhoea. In extreme circumstances, occlusion of the intestines may occur with *A. galli*, leading to death (Taylor et al., 2007). Control (treatment) of these and other parasites, especially in small holder, free range and organic farming enterprises, remains an ongoing requirement in chicken production in the United Kingdom and indeed worldwide. The increasing popularity of unlicensed herbal based products, is driven in part by concerns about conventional anthelmintic use, including the potential for drug resistance and possible drug residues in meat and or eggs.

A comparative efficacy study in chickens was undertaken in May 2010 to evaluate the efficacy of a popular, commercially available herbal product containing sunflower oil, seaweed meal, dicalcium phosphate, *Allium sativum*, *Cinnamomum zelandicum*, *Mentha piperita*, *Thymus vulgaris*, *Galium aperine* and *Capsicum minimum* and a conventional commercially available flubendazole-based anthelmintic product against all three targeted parasites. Both products claim efficacy against internal parasites of

* Corresponding author. Tel.: +44 1594 530 809; fax: +44 1594530767.
E-mail address: sarah@ridgewayresearch.co.uk (S. Squires).

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chickens. The flubendazole (Flubenvet®, Janssen Animal Health) label states 'Flubenvet® is effective against gapeworm, large roundworm, caecal worm, hairworm, and gizzard worm in chickens, turkeys, and geese. Treatment over seven days in feed. 'The herbal product (Verm-X® Paddocks Farm Partnerships) label states 'Verm-X® can control all known internal parasites. Give in addition to normal feed for three days every month'. On productivity and welfare grounds and to avoid the possible development of resistance there is a need to ensure that all products claiming anthelmintic efficacy deliver acceptable efficacy.

2. Materials and methods

A total of 48 laying hens were selected from a certified organic farm in Hertfordshire, on the basis of previous faecal egg count data and post mortem examination of other chickens from within the flock which indicated that all of the chickens had a worm burden. None of the 48 birds had ever been treated with an anthelmintic or any alternative treatment targeted at internal parasites. The animals were transported to the study site and allowed at least seven days to acclimatise to the study environment prior to treatment, during which time they were fed on layers mash.

On Day 0 the birds were randomised on the basis of FEC and bodyweight into one of three treatment groups each of 16 birds:

Group A: flubendazole-treated group (30 ppm). On Day 0 the feed for Group A was replaced with flubendazole medicated feed until Day 6 inclusive (seven days of treatment).
Group B: herbal product-treated group (2.5 g/bird/day). Group B birds received the herbal based product (as per manufacturer's instructions) added to the base of their feeders under the layers mash for three consecutive days (Day 0–Day 2). The feeders and catcher trays were examined to confirm consumption of the herbal pellets.
Group C: control (no-treatment) group. Group C birds remained on layers mash throughout the study.

Each group was randomly allocated to a pen. Prior to treatment on Day 0, the bird with the median weight from each group was euthanased to confirm the parasite species present. All three euthanased birds had adult *A. galli*, *H. gallinarum* and *Capillaria* spp. present. Following the periods of treatment, Groups A and B were fed layers mash until Day 12 when feed was removed from all study pens. Feed consumption was measured by weighing feed remaining and waste feed at time points throughout the study for each pen.

On Day 12 the birds were euthanased by cervical dislocation. A faecal sample was taken from each bird *post mortem*. Autopsies were performed on each bird to remove the small intestine and caecae. The small intestine and caecae were opened using scissors and any visibly present worms were removed. The contents of the small intestine and the caecae were washed through a 53 μm sieve. The worms found on the surface of the sieve were washed into individually labelled pots and iodine was added to preserve the worms. Worms were counted and speciated according to Ridgeway Research standard operating procedures. Fourth stage larvae (L4) were identified and counted. Fifth stage larvae (L5) were classed as adults.

FECs were performed using the Modified McMaster Technique over days 14 and 15 (sensitivity 50 eggs per gram) (Thienpont et al., 1979). *H. gallinarum* and *A. galli* eggs were not differentiated and were counted together.

Statistical tests were performed on the worm count data (primary parameter) and the egg count data (secondary parameter). Tests were performed using SAS statistical program (SAS/STAT User's Version 9.1, SAS Institute, Cary, NC). Efficacy (based on worm counts) was assessed using the following equation:

$$\%\text{Efficacy (reduction)} = \left[\frac{N2 - N1}{N2}\right] \times 100$$

where $N1$ = Geometric mean count of species for the group treated with flubendazole or the herbal product. $N2$ = Geometric mean count of species for the no-treatment control group (Group C)

3. Results

Table 1 gives a summary of the worm counts. Total worm counts and speciated worm counts were analysed using ANOVA and post hoc tests (Dunnett's and Tukey's).

Table 1
Summary of worm counts including means, efficacy and number of positive animals in each group.

Group (n = 15)		No. of animals positive	Worm counts Mean	Std. dev.	Min	Max	% Efficacy
A	Total worm count	5	5.6*	7.8	1.0	19.0	99.4
Flubendazole	*A. galli*	0	0*	0	0	0	100.0
	Capillaria spp.	1	2.0*		2.0	2.0	97.8
	H. gallinarum	4	6.3*	8.6	1.0	19.0	99.2
B	Total worm count	15	534.9 NS	383.8	108.0	1327.0	−50.7
Herbal	*A. galli*	15	10.0 NS	6.3	2.0	21.0	11.6
	Capillaria spp.	15	130.7 NS	86.9	14.0	330.0	−110.2
	H. gallinarum	15	394.1 NS	342.5	49.0	1198.0	−47.2
C	Total worm count	15	355.3	258.0	74.0	958.0	n/a
Untreated control	*A. galli*	15	11.3	6.5	2.0	23.0	
	Capillaria spp.	15	99.8	139.4	10.0	374.0	
	H. gallinarum	15	244.3	167.2	47.0	585.0	

* Significantly different ($p < 0.05$) from untreated controls (Group C).
NS: not significantly different ($p < 0.05$) from untreated controls (Group C).

Group A total worm counts and species worm counts were significantly different ($p < 0.05$) from Group B or C. Groups B and C (untreated control) total and species worm counts were not significantly different from each other.

All animals in the control group showed adequate infections (based on VICH GL21 guidelines).

Post treatment faecal egg count data revealed no significant differences between any of the groups.

Feed consumption analysis (Kruskal–Wallis test) showed no significant differences between groups. Body weight comparisons for all groups pre and post treatment showed no significant differences.

Efficacy was calculated for Groups A and B. The efficacy results are shown in Table 1.

4. Discussion and conclusions

This study demonstrated that with chickens originating from an organic farm where parasitism was well established, a flubendazole based product was efficacious against targeted parasites while the herbal based product was not efficacious. Both products were administered at the labelled doses for the labelled durations and neither product label indicated that repeated therapy was required to deliver adequate efficacy. The World Association for the Advancement of Veterinary Parasitology (WAAVP) guidelines (Yazwinski et al., 2003) state that efficacy can be claimed if there is a ≥90% reduction in worm count compared to untreated controls. Worm counts were the primary efficacy parameter assessed in this study. On this basis, the herbal product was not efficacious in the treatment of *A. galli*, *H. gallinarum* or *Capillaria* spp (no significant difference compared to untreated controls). In this study no significant differences were seen in the mean faecal egg counts of any of the three groups. FEC's are indicative of efficacy in live birds but are recognised as being highly variable between birds and between stool samples from the same bird, depending on parasite egg shedding and as such the standard method for quantifying efficacy is by direct worm counts at necropsy (Yazwinski et al., 2003).

At present, commercially available herbal products claiming anthelmintic properties fall outside the remit of the veterinary medicinal products Directive 2004/28/EC (Article 17 and 33–38) and as such do not require a licence and can be freely marketed without demonstrating specific anthelmintic efficacy. Such products are a concern from many aspects but most importantly from an animal welfare perspective – sick birds with diarrhoea if burdens build up. Ineffective anthelmintic treatment of parasitized birds can also have significant longer term economic consequences for the farmer via reduced weight gain, ill-thrift and possibly reduced egg production in the flock as well as the direct costs associated with purchasing and administering an ineffective product. With limited regulatory and veterinary intervention in the supply and use of herbal product ineffective treatments will lead to continuing welfare issues emerging in the future. More comparative efficacy studies, similar to this one, are therefore recommended to generate data on the efficacy or otherwise of these products.

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